

Priyansh Verma

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Professional Summary

B.Tech + M.Tech Dual Degree CSE + AI at IIT Bombay. CGPA 9.6/10. Published ML researcher (NeurIPS 2023 workshop). Contributor to open-source LLM tooling. Interned at Google DeepMind and Microsoft Research.

Technical Skills

Languages:	Python, C++, CUDA, Julia
ML/DL:	PyTorch, JAX, TensorFlow, TensorRT, DeepSpeed, ONNX
LLM/NLP:	HuggingFace Transformers, LLaMA, BERT, T5, LoRA, PEFT, LangChain, RAG
Computer Vision:	OpenCV, YOLOv8, Detectron2, CLIP, Stable Diffusion
MLOps:	Kubernetes, Docker, Ray, vLLM, MLflow, DVC, Weights and Biases, GCP Vertex AI

Key Projects

LLM Fine-Tuning Pipeline -- LoRA + QLoRA (PyTorch + DeepSpeed)

Production fine-tuning for 7B/13B LLaMA models. 4-bit QLoRA reduced GPU memory by 60%. DeepSpeed ZeRO-3 across 8xA100 GPUs. MMLU benchmark 72.3% vs 67.1% base. Published at NeurIPS 2023 workshop.

Technologies: PyTorch, DeepSpeed, LoRA, QLoRA, LLaMA, HuggingFace PEFT, Weights and Biases, CUDA

Multimodal RAG System -- CLIP + LangChain + pgvector

Enterprise document QA combining visual and textual retrieval. CLIP embeddings + sentence-transformers stored in pgvector. GPT-4 for answer synthesis. 95% retrieval accuracy. Deployed on GCP with Kubernetes.

Technologies: CLIP, LangChain, pgvector, GPT-4, SentenceTransformers, FastAPI, GCP, Kubernetes

Real-Time Object Detection -- YOLOv8 + TensorRT (Edge AI)

YOLOv8 optimized with TensorRT for NVIDIA Jetson Orin. 78 FPS vs 12 FPS baseline. Deployed across 200 smart cameras for warehouse automation at Reliance Retail.

Technologies: YOLOv8, TensorRT, ONNX, Python, OpenCV, CUDA, NVIDIA Jetson, Docker

Education

B.Tech + M.Tech Dual Degree, CSE + AI, IIT Bombay, 2020-2025. CGPA: 9.6/10. Publications: NeurIPS 2023, ICLR 2024 (workshops).